

1.4 The Matrix Equation $Ax=b$

Definition:

If A is an $m \times n$ matrix, with columns $a_1, \dots, a_n$, and if x is in $\mathbb{R}^n$, then the product of A and x , denoted by Ax , is the linear combination of the columns of A using the corresponding entries in x as weights; that is

$$Ax = [a_1 \ a_2 \ \dots \ a_n] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 a_1 + x_2 a_2 + \dots + x_n a_n$$

Note that Ax is defined only if the number of columns of A equals the number of entries in x .

Example 1:

$$\text{a) } \begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ -1 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ -5 \end{bmatrix} + 7 \begin{bmatrix} -1 \\ 3 \end{bmatrix} \quad (\text{From defn.})$$
$$= \begin{bmatrix} 4 \\ 0 \end{bmatrix} + \begin{bmatrix} 6 \\ -15 \end{bmatrix} + \begin{bmatrix} -7 \\ 21 \end{bmatrix} \quad (\text{Scalar multiplication})$$